

LOW ACRYLONITRILE NBR PATENT LANDSCAPE: COMPOSITION CONTROL, HYDROGENATION, AND HIGH-PERFORMANCE RUBBER FORMULATIONS

1. ABSTRACT

This report synthesizes four primary patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile elastomer systems. The target is interpreted as NBR compositions with reduced acrylonitrile content relative to conventional grades, while preserving oil resistance, cold resistance, processability, and mechanical performance. Across the selected patents, the disclosed chemistry spans both direct NBR copolymerization and post-polymerization hydrogenation to highly saturated nitrile rubbers (HSNBR/HNBR-like materials), with acrylonitrile contents explicitly ranging from about 14 wt% to 36 wt% in the examples and claims. The landscape shows two major formulation directions: (1) composition tuning of the rubber backbone through monomer-feed control and hydrogenation to achieve a desired nitrile content and iodine value, and (2) reinforcement/crosslinking packages that compensate for lower nitrile content or altered saturation by using silica, zinc dimethacrylate, fillers, plasticizers, polyamine crosslinkers, and in one case cellulose nanofiber or powder super absorbent polymer additives.

EP3187530B1 emphasizes highly saturated nitrile rubber containing 5 to 50 wt% nitrile monomer units and a low-iodine backbone, then reinforces it with cellulose nanofiber master batches and polyamine crosslinking. US8133948B2 focuses on highly saturated nitrile rubber with 10 to 60 wt% nitrile units, especially 15 to 55 wt%, combined with metallic carboxylate salts and treated silica to improve heat resistance and tensile stress. EP3351591B1 discloses a family of nitrile-group-containing highly saturated copolymer rubbers with 14 to 20.5 wt% nitrile units in the composition claims, prepared by emulsion polymerization followed by hydrogenation, then formulated with fillers and plasticizers for cold/oil resistance. US10385647B2 is not a synthesis patent for NBR, but it is relevant as a formulation patent showing that nitrile elastomers, including hydrogenated nitrile elastomers, can be used with powder SAPs to achieve high swell performance under brine and high-temperature conditions. Overall, the evidence supports low-ACN NBR design through precise monomer ratio control, hydrogenation level control, and downstream compounding choices rather than a single synthesis route.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3187530B1
- Patent Title: Crosslinkable nitrile rubber composition and crosslinked rubber material
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of a highly saturated nitrile rubber latex by emulsion polymerization of acrylonitrile, 1,3-butadiene, mono-n-butyl maleate, and t-dodecyl mercaptan in water with sodium dodecylbenzene sulfonate, followed by hydrogenation using a palladium catalyst in an autoclave. The resulting latex is coagulated and dried, then compounded with cellulose nanofiber master batch, carbon black, plasticizer, antioxidant, polyamine crosslinker, and DBU accelerator to form a cross-linkable nitrile rubber composition.

- Ion exchanged water: 180 parts — Synthesis Example 1 - Sodium dodecylbenzene sulfonate aqueous solution (10 wt%): 25 parts — Synthesis Example 1 - Acrylonitrile: 37 parts — Synthesis Example 1 - Mono-n-butyl maleate: 6 parts — Synthesis Example 1 - t-Dodecyl mercaptan: 0.5 part — Synthesis Example 1 - 1,3-Butadiene: 57 parts — Synthesis Example 1 - Polymerization temperature: 5°C — Synthesis Example 1 - Cumen hydroperoxide: 0.1 part — Synthesis Example 1 - Hydroquinone aqueous solution (10 wt%): 0.1 part — Synthesis Example 1 - Polymerization addition rate: 82% — Synthesis Example 1 - Residual monomer removal temperature: 60°C — Synthesis Example 1 - Palladium loading: 1,000 ppm by weight relative to dry rubber — Synthesis Example 1 - Hydrogen pressure: 3 MPa — Synthesis Example 1 - Hydrogenation temperature: 50°C — Synthesis Example 1 - Hydrogenation time: 6 hours — Synthesis Example 1 - Latex solid content concentration: about 30 wt% — Synthesis Example 1 - Volume average particle size: 100 nm — Synthesis Example 1 - Acrylonitrile units: 35.6 wt% — Synthesis Example 1 - Butadiene units: 59.0 wt% — Synthesis Example 1 - Mono-n-butyl maleate units: 5.4 wt% — Synthesis Example 1 - Iodine value: 8 — Synthesis Example 1 - Carboxyl group content: 3.1×10^{-2} ephr — Synthesis Example 1 - Polymer Mooney viscosity [ML1+4, 100°C]: 48 — Synthesis Example 1 - Methyl ethyl ketone insolubles: 1.1 wt% — Synthesis Example 1 - Cellulose nanofiber solid content concentration: 10 wt% — Example 1 - Highly saturated nitrile rubber to cellulose nanofiber ratio: 100:10 weight ratio — Example 1 - Sodium chloride coagulating solution: 10 wt% — Example 1 - Sodium chloride amount: 100 parts with respect to 100 parts of highly saturated nitrile rubber — Example 1 - Master batch (MB1): 110 parts — Example 1 - N550 carbon black: 40 parts — Example 1 -

4,4-di-(α,α -dimethylbenzyl)diphenylamine: 1.5 parts — Example 1 - Tri-2-ethylhexyl trimellitate: 5 parts — Example 1 - Stearic acid: 1 part — Example 1 - Polyoxyethylenealkyl ether phosphoric acid ester: 1 part — Example 1 - 2,2-bis[4-(4-aminophenoxy)phenyl]propane: 6.3 parts — Example 1 - 1,8-diazabicyclo[5,4,0]undecene-7 60% product: 4 parts — Example 1 - Cellulose nanofiber ratio: 100:15 — Example 2 - Master batch (MB2): 115 parts — Example 2 - Magnesium sulfate aqueous solution: 5 wt% — Example 3 - Magnesium sulfate amount: 20 parts with respect to 100 parts of highly saturated nitrile rubber — Example 3 - Master batch (MB3): 110 parts — Example 3 - Cellulose fiber average fiber diameter: 15 μm — Comparative Example 2 - Cellulose fiber solid content concentration: 10 wt% — Comparative Example 2 - Master batch (MB4): 110 parts — Comparative Example 2

Relevant Experimental Evidence - Synthesis Example 1: Demonstrated emulsion polymerization followed by Pd-catalyzed hydrogenation to a highly saturated nitrile rubber with 35.6 wt% acrylonitrile units and iodine value 8. - Example 1: Showed that a cellulose nanofiber master batch combined with crosslinking agents produced a cross-linkable nitrile rubber composition with good tensile performance. - Example 2: Increasing cellulose nanofiber content from 10 to 15 parts was feasible and still yielded a workable composition. - Example 3: Replacing sodium chloride with magnesium sulfate as coagulant still produced a usable master batch and composition. - Comparative Example 2: Larger cellulose fiber (15 μm diameter) was inferior to nanofiber in the reinforced composition.

Technical Relevance This patent is directly relevant because it discloses a nitrile rubber backbone with explicit acrylonitrile content and a hydrogenated, low-iodine structure, then shows how reinforcement and crosslinking can preserve performance. It provides a practical route for low-ACN nitrile rubber design through monomer-feed control and post-polymerization hydrogenation.

Patent 2

- Patent Number: US8133948B2
- Patent Title: Nitrile rubber composition, cross-linkable nitrile rubber composition and cross ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile rubber compositions based on highly saturated nitrile rubber having an iodine value of 100 or less, compounded with α,β -ethylenically unsaturated carboxylic acid metallic salts and treated silica, then crosslinked with an organic peroxide. The rubber is described as being obtainable by copolymerization of nitrile monomer with diene and/or α -olefin monomers by emulsion, suspension, bulk, or solution polymerization, with emulsion polymerization preferred. - Iodine

value: 100 or less — Abstract/Claims - α,β -Ethylenically unsaturated nitrile monomer unit content: 10 to 60 wt% — Description - α,β -Ethylenically unsaturated nitrile monomer unit content: 15 to 55 wt% — Description - α,β -Ethylenically unsaturated nitrile monomer unit content: 20 to 50 wt% — Description - Diene monomer unit and/or α -olefin monomer unit content: 20 to 90 wt% — Description - Diene monomer unit and/or α -olefin monomer unit content: 30 to 85 wt% — Description - Diene monomer unit and/or α -olefin monomer unit content: 40 to 80 wt% — Description - Diene monomer unit and/or α -olefin monomer unit content: 50 to 80 wt% — Description - α,β -Ethylenically unsaturated carboxylic acid metallic salt content: 3 to 120 parts by weight per 100 parts by weight of nitrile rubber — Claims - Calcinated wet silica content: 5 to 300 parts by weight per 100 parts by weight of nitrile rubber — Claims - Crosslinking agent content: 0.2 to 20 parts by weight per 100 parts by weight of nitrile rubber — Description - Silica (1) specific surface area: 110 m²/g — Example 1 note - Silica (1) pH: 3.8 — Example 1 note - Example 1 kneading temperature: 50°C — Example 1 - Zinc dimethacrylate: 15 parts — Example 1 - 4,4'-di-(α,α' -dimethylbenzil)diphenylamine: 1.5 parts — Example 1 - Zinc salt of 2-mercaptobenzimidazole: 1.5 parts — Example 1 - Silica (1): 40 parts — Example 1 - Zinc oxide: 10 parts — Example 1 - Trimellitic acid ester: 8 parts — Example 1 - Organic peroxide: 8 parts of 40% product (3.2 parts pure peroxide) — Example 1 - Silica (2) specific surface area: 133 m²/g — Example 2 note - Silica (2) pH: 6.9 — Example 2 note - Silica (3) specific surface area: 195 m²/g — Comparative Example 1 note - Silane coupling agent: 2 parts — Comparative Example 2 - Silica (4) specific surface area: 200 m²/g — Comparative Example 3 note

Relevant Experimental Evidence - Example 1: Silane-treated fumed silica with zinc dimethacrylate gave a cross-linkable nitrile rubber composition with strong heat-aging performance. - Example 2: Calcinated wet silica also produced a composition with good tensile properties and heat resistance. - Comparative Example 1: Wet silica without the preferred treatment reduced heat-aging elongation performance. - Comparative Example 2: Adding silane coupling agent to wet silica did not recover the heat resistance advantage. - Comparative Example 4: Omitting zinc dimethacrylate reduced tensile stress, showing its importance in the formulation.

Technical Relevance This patent is relevant because it defines a highly saturated nitrile rubber backbone with explicit nitrile content ranges and shows how filler/crosslink packages affect final properties. The disclosed composition window is useful for designing low-ACN nitrile rubbers that still retain oil and heat resistance.

Patent 3

- Patent Number: EP3351591B1
- Patent Title: Nitrile rubber composition, cross-linkable nitrile rubber composition, and ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses emulsion polymerization of acrylonitrile, mono-n-butyl maleate, a monocarboxylic acid ester monomer, and 1,3-butadiene in water with sodium dodecylbenzenesulfonate and t-dodecyl mercaptan, followed by hydrogenation with a palladium catalyst. The hydrogenated latex is coagulated with methanol and dried to obtain nitrile-group-containing highly saturated copolymer rubbers with varying acrylonitrile contents, which are then compounded into nitrile rubber compositions with fillers, plasticizers, and crosslinking agents. - Ion exchanged water: 180 parts — Synthesis Example 1 - Sodium dodecylbenzenesulfonate aqueous solution (10 wt%): 25 parts — Synthesis Example 1 - Acrylonitrile: 15 parts — Synthesis Example 1 - Mono-n-butyl maleate: 6 parts — Synthesis Example 1 - n-Butyl acrylate: 39 parts — Synthesis Example 1 - t-Dodecyl mercaptan: 0.5 part — Synthesis Example 1 - 1,3-Butadiene: 40 parts — Synthesis Example 1 - Polymerization temperature: 5°C — Synthesis Example 1 - Cumene hydroperoxide: 0.1 part — Synthesis Example 1 - Hydroquinone aqueous solution (10 wt%): 0.1 part — Synthesis Example 1 - Polymerization time: 16 hours — Synthesis Example 1 - Residual monomer removal temperature: 60°C — Synthesis Example 1 - Palladium loading: 1,000 ppm by weight relative to dry rubber — Synthesis Example 1 - Hydrogen pressure: 3.0 MPa — Synthesis Example 1 - Hydrogenation temperature: 50°C — Synthesis Example 1 - Hydrogenation time: 6 hours — Synthesis Example 1 - Methanol: two times volume — Synthesis Example 1 - Drying temperature: 60°C — Synthesis Example 1 - Drying time: 12 hours — Synthesis Example 1 - Acrylonitrile units: 15 wt% — Synthesis Example 1 - Mono-n-butyl maleate units: 5 wt% — Synthesis Example 1 - n-Butyl acrylate units: 35 wt% — Synthesis Example 1 - 1,3-Butadiene units: 45 wt% — Synthesis Example 1 - Iodine value: 10 — Synthesis Example 1 - Carboxyl group content: 2.8×10^{-2} ephr — Synthesis Example 1 - Polymer Mooney viscosity [ML1+4, 100°C]: 40 — Synthesis Example 1 - Acrylonitrile used: 21 parts — Synthesis Example 2 - Mono-n-butyl maleate used: 6 parts — Synthesis Example 2 - n-Butyl acrylate used: 34 parts — Synthesis Example 2 - 1,3-Butadiene used: 39 parts — Synthesis Example 2 - Acrylonitrile units: 21 wt% — Synthesis Example 2 - n-Butyl acrylate units: 29 wt% — Synthesis Example 2 - Polymer Mooney viscosity [ML1+4, 100°C]: 47 — Synthesis Example 2 - 2-Methoxyethyl acrylate used: 30 parts — Synthesis Example 3 - Acrylonitrile used: 24 parts — Synthesis Example 3 - Mono-n-butyl maleate used: 7 parts — Synthesis Example 3 - 1,3-Butadiene used: 39 parts — Synthesis Example 3 - Acrylonitrile units: 24 wt% — Synthesis Example 3 - Mono-n-butyl maleate units: 6 wt% — Synthesis Example 3 - 2-Methoxyethyl acrylate units: 25 wt% — Synthesis Example 3 - Polymer Mooney viscosity [ML1+4, 100°C]: 48 — Synthesis Example 3 - Acrylonitrile used: 16.5 parts — Synthesis Example 4 - Mono-n-butyl maleate used: 6 parts — Synthesis Example 4 - n-Butyl acrylate used: 37.8 parts — Synthesis Example 4 - 1,3-Butadiene used: 39.8 parts — Synthesis Example 4 - Acrylonitrile units: 16.5 wt% — Synthesis Example 4 - n-Butyl acrylate units: 33.5 wt% — Synthesis Example 4 - Polymer Mooney viscosity [ML1+4, 100°C]: 45 — Synthesis Example 4 - Acrylonitrile used: 18.0 parts — Synthesis Example 5 - Mono-n-butyl maleate used: 6 parts — Synthesis Example 5 - n-Butyl acrylate used: 36.5 parts — Synthesis Example 5 - 1,3-Butadiene used: 39.5 parts — Synthesis Example 5 - Acrylonitrile units: 18 wt% — Synthesis Example 5 - n-Butyl acrylate units: 32 wt% — Synthesis Example 5 - Polymer Mooney viscosity [ML1+4, 100°C]: 43 — Synthesis Example 5 - Acrylonitrile used: 19.5 parts — Synthesis Example 6 - Mono-n-butyl maleate used: 6 parts — Synthesis Example 6 - n-Butyl acrylate used: 35.3 parts — Synthesis Example 6 - 1,3-Butadiene used: 39.3 parts — Synthesis Example 6 - Acrylonitrile units:

19.5 wt% — Synthesis Example 6 - n-Butyl acrylate units: 30.5 wt% — Synthesis Example 6 - Polymer Mooney viscosity [ML1+4, 100°C]: 42 — Synthesis Example 6 - Palladium loading: 900 ppm by weight — Synthesis Example 7 - Iodine value: 25 — Synthesis Example 7 - Palladium loading: 800 ppm by weight — Synthesis Example 8 - Iodine value: 45 — Synthesis Example 8 - Palladium loading: 650 ppm by weight — Synthesis Example 9 - Iodine value: 75 — Synthesis Example 9 - Silica: 70 parts — Example 1 - Polyether ester-based plasticizer: 20 parts — Example 1 - Stearic acid: 1 part — Example 1 - 4,4'-di-(α,α' -dimethylbenzyl)diphenylamine: 1.5 parts — Example 1 - Polyoxyethylene alkyl ether phosphoric acid ester: amount not numerically specified — Example 1 - DBU: 4 parts — Example 1 - Hexamethylene diamine carbamate: 2 parts — Example 1 - A1 content in Example 2 blend: 25 wt% — Example 2 - A2 content in Example 2 blend: 75 wt% — Example 2

Relevant Experimental Evidence - Synthesis Example 1: Produced a hydrogenated nitrile-group-containing copolymer with 15 wt% acrylonitrile units and iodine value 10. - Synthesis Examples 2 to 6: Demonstrated composition tuning from 16.5 to 24 wt% acrylonitrile while maintaining hydrogenated backbones and similar carboxyl content. - Synthesis Examples 7 to 9: Showed that lowering palladium loading to 900, 800, and 650 ppm increased iodine value to 25, 45, and 75, respectively. - Example 1: Formulated a cross-linkable nitrile rubber composition using 70 parts silica, 20 parts plasticizer, DBU, and hexamethylene diamine carbamate. - Example 2: Demonstrated blending two different hydrogenated nitrile rubbers (A1 and A2) at 25:75 to tailor composition.

Technical Relevance This patent is highly relevant because it explicitly targets low-acrylonitrile, highly saturated nitrile copolymers and provides a broad composition window centered around 14 to 20.5 wt% nitrile units. It also shows how hydrogenation level and compounding ingredients can be tuned to balance cold resistance, oil resistance, and processability.

Patent 4

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses swellable rubber compositions for downhole packers rather than NBR synthesis. The compositions include a nitrile elastomer, butadiene rubber, sulfur curing agent, silica, and a powder super absorbent polymer, with examples showing curing and swelling performance in brine at elevated temperature and pressure. - Powder super absorbent polymer: about 125 Phr to about 175 Phr — Claims - Powder super absorbent polymer: about 170 Phr to about 210 Phr — Claims - Powder super absorbent

polymer: at least about 150 Phr, 175 Phr, 200 Phr or 225 Phr — Claims - Hydrogenated nitrile elastomer percent hydrogenation: about 70 percent to about 91 percent — Description - Elastomer Mooney viscosity ML1+4 (100°C): about 47 to about 72 — Description - Hydrogenated nitrile elastomer percent unsaturation: about 9 percent to about 30 percent — Description - Swelling in 5% brine at 200°F: at least about 50% to 75% by volume in about 96 hours — Description - Swelling in 5% brine at 275°F: at least about 50% to 75% by volume in about 96 hours — Description - Pressure capability: at least about 3000, 4000, 5000 or 6000 psi — Description - Base pipe outer diameter: 4.5 inches — Example 1 - Packer outer diameter: 5.7 inches — Example 1 - Packer length: 5 feet — Example 1 - Curing temperature: 330°F — Example 1 - Curing pressure: 70 psi — Example 1 - Curing time: 2.5 hours — Example 1 - Brine concentration: 16% — Example 1 - Test temperature: 275°F — Example 1 - Test pressure: about 300 psi — Example 1 - Differential pressure requirement: at least 5000 psi — Example 1 - Hardness: about 86 to 92 Shore A — Example 1 - SAP loading: 110 phr — Example 2 - SAP loading: 130 phr — Example 2 - SAP loading: 150 phr — Example 2 - SAP loading: 170 phr — Example 2 - Powder SAP mean particle size: 35.33 μm — Example 2 - Powder SAP median particle size: 5 μm — Example 2 - Powder SAP standard deviation: 10.61 μm — Example 2 - Medium granular SAP mean particle size: 298.67 μm — Example 2 - Medium granular SAP median particle size: 261 μm — Example 2 - Medium granular SAP standard deviation: 97.72 μm — Example 2 - Large granular SAP mean particle size: 666.23 μm — Example 2 - Large granular SAP median particle size: 652 μm — Example 2 - Large granular SAP standard deviation: 161.09 μm — Example 2 - 200°F test temperature — Example 2 - 275°F test temperature — Example 2 - 5% brine by volume — Example 2 - 16% brine by volume — Example 2 - Predicted powder SAP loading for 75% swell: about 205 Phr — Example 3 - Predicted medium granular SAP loading for 75% swell: about 190 Phr — Example 3 - Maximum medium granular SAP load: about 170 Phr — Example 3 - Predicted large granular SAP loading for 75% swell: about 160 Phr — Example 3 - Maximum large granular SAP load: about 150 Phr — Example 3 - Composition containing 10% NBR rubber (Nipol DN3350) — Example 4 - Composition containing 10% HNBR (Zetpol 2030L) — Example 4 - 96 hour swell at 200°F, 5% brine: 75.8% for 10% DN3350 — Example 4 - 96 hour swell at 200°F, 5% brine: 77.8% for 10% HNBR — Example 4 - 96 hour swell at 200°F, 16% brine: 46.5% for 10% DN3350 — Example 4 - 96 hour swell at 200°F, 16% brine: 47.2% for 10% HNBR — Example 4 - 96 hour swell at 275°F, 5% brine: 98.4% for 10% DN3350 — Example 4 - 96 hour swell at 275°F, 5% brine: 96.0% for 10% HNBR — Example 4 - 96 hour swell at 275°F, 16% brine: 65.4% for 10% DN3350 — Example 4 - 96 hour swell at 275°F, 16% brine: 66.9% for 10% HNBR — Example 4

Relevant Experimental Evidence - Example 1: A packer using Composition B was cured at 330°F and 70 psi for 2.5 hours and then held in 16% brine at 275°F, showing hardness of about 86 to 92 Shore A and no bonding issues. - Example 2: Powder, medium granular, and large granular SAPs were compared at 110 to 170 phr, showing strong dependence of swell and leaching on particle size and loading. - Example 3: Predicted 75% swell loading was about 205 phr for powder SAP, higher than the practical limits for granular SAPs. - Example 4: A 10% NBR composition and a 10% HNBR composition both showed strong swell performance, with HNBR offering slightly higher swell in some brine conditions.

Technical Relevance Although not a synthesis patent, this disclosure is relevant because it demonstrates how nitrile elastomers, including hydrogenated nitrile elastomers, are formulated for swellable packers under harsh brine and temperature conditions. It provides useful compounding context for low-ACN nitrile rubber applications where swelling, leaching, and mechanical integrity are critical.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP3187530B1 and EP3351591B1 both use emulsion polymerization followed by Pd-catalyzed hydrogenation, but EP3187530B1 centers on a highly saturated nitrile rubber with 35.6 wt% acrylonitrile units in the example, whereas EP3351591B1 explicitly targets lower nitrile contents such as 15 to 24 wt% in the synthesis examples and claims 14 to 20.5 wt%.
- US8133948B2 and EP3351591B1 both define highly saturated nitrile rubbers with iodine value limits, but US8133948B2 is broader on nitrile content (10 to 60 wt%) and emphasizes silica/zinc dimethacrylate reinforcement, while EP3351591B1 emphasizes composition control and blending of different rubbers to tune cold/oil resistance.
- EP3187530B1 uses cellulose nanofiber master batches and a polyamine crosslinker (2,2-bis[4-(4-aminophenoxy)phenyl]propane) with DBU, whereas US8133948B2 uses zinc dimethacrylate, silica, zinc oxide, and organic peroxide crosslinking.
- Hydrogenation conditions are similar in EP3187530B1 and EP3351591B1: both use 3 MPa hydrogen pressure, 50°C, and 6 hours with palladium catalyst, but EP3351591B1 varies palladium loading from 1,000 to 650 ppm to modulate iodine value from 10 to 75.
- US10385647B2 is formulation-focused rather than synthesis-focused; it uses nitrile elastomers, including hydrogenated nitrile elastomers with 70 to 91% hydrogenation, together with powder SAPs at 125 to 210 phr to achieve swell performance in brine, which is orthogonal but relevant to end-use compounding of low-ACN nitrile systems.
- Across the patents, lower acrylonitrile content is associated with improved cold resistance but requires compensating formulation strategies: reinforcement fibers in EP3187530B1, silica/zinc dimethacrylate in US8133948B2, and filler/plasticizer balancing in EP3351591B1.
- Coagulation and isolation methods vary: EP3187530B1 uses sodium chloride or magnesium sulfate coagulation of latex master batches, while EP3351591B1 uses methanol coagulation and vacuum drying after hydrogenation.

4. CONCLUSION

The patent evidence shows that low-acrylonitrile NBR and related highly saturated nitrile rubbers are best approached through controlled emulsion copolymerization, followed where needed by hydrogenation to reduce unsaturation while preserving nitrile functionality. The most directly relevant composition window is EP3351591B1, which explicitly centers on 14 to 20.5 wt% nitrile

units and demonstrates practical synthesis examples at 15 to 24 wt% acrylonitrile. EP3187530B1 and US8133948B2 support the broader design space for highly saturated nitrile rubbers, showing that performance can be recovered through reinforcement and crosslinking packages even when nitrile content is adjusted. For low-ACN synthesis, the evidence favors emulsion polymerization at about 5°C with controlled monomer feed, mercaptan chain transfer, and Pd hydrogenation at 3 MPa and 50°C for 6 hours, with nitrile content tuned near 15 to 20 wt% where cold resistance is prioritized. For end-use formulations, silica, zinc dimethacrylate, plasticizers, and appropriate crosslinking systems are important to maintain tensile, heat, and oil resistance.

5. REFERENCES

- EP3187530B1 | Crosslinkable nitrile rubber composition and crosslinked rubber material | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3187530B1/en>
- US8133948B2 | Nitrile rubber composition, cross-linkable nitrile rubber composition and cross ... | Not disclosed | US | 2012 | <https://patents.google.com/patent/US8133948B2/en>
- EP3351591B1 | Nitrile rubber composition, cross-linkable nitrile rubber composition, and ... | Not disclosed | EP | 2020 | <https://patents.google.com/patent/EP3351591B1/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>